CS-GY 6513-C Big Data Project Report

Spring 2026

U.S. Airline Flight Delay Analysis

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1. Executive summary:

Project Name: U.S. Airline Flight Delay Analysis

Brief summary: This project analyzes large-scale U.S. airline on-time performance data from the 2024 BTS dataset to understand how flight delays behave in real-world operations.The dataset consists of 12 monthly ZIP files and contains millions of flight records (over 3GB of data). To efficiently process this large-scale dataset, we used PySpark and Spark SQL for distributed data processing, data cleaning, and analytical queries.

The project focuses on identifying delay patterns and uncovering key factors that influence flight delays across different dimensions, including airlines, airports, time, and operational conditions. Through data analysis and visualization, we discovered important patterns such as delay accumulation throughout the day, congestion effects at high-traffic airports, and variation across airlines. In addition, we implemented a machine learning model to predict flight delays. However, the model achieved only moderate accuracy (around 60–78%) due to missing critical features such as weather conditions and real-time operational data, highlighting the limitations of predictive modeling in this context.

The main **objectives** of this project are:

- To analyze delay patterns across airlines, airports, and time
- To identify key factors influencing flight delays
- To study how delays vary across different conditions (time of day, season, distance, etc.)
- To discover patterns such as delay propagation and congestion effects
- To build a machine learning model for delay prediction and evaluate its limitations

Technologies used include:

- PySpark – large-scale data processing and feature engineering
- Spark SQL – analytical querying and aggregation
- Matplotlib – visualization of delay patterns and trends
- Spark ML (Logistic Regression) – predictive modeling
- Google Colab – cloud-based development environment

2. Code Execution Instructions:

Link to the git repository: https://github.com/TianqiZhen426/Big-Data_2026Spring_Project

The readme provides all steps to reproduce the results using Google Colab:

README Link:

https://github.com/TianqiZhen426/Big-Data_2026Spring_Project/blob/main/README.md

Steps to Run the Code:

1. First, go to Google Drive, and in My Drive, create a new folder named 'BigDataProject'.
2. Download the dataset from

https://drive.google.com/file/d/1iXESKbrhonsKsb8BJKz7R4o4HPaEYRPj/view?usp=sharing

(the file BTS_Data_2024.zip).

After downloading, upload this ZIP file into the ‘BigDataProject’ folder in your Google Drive.

3. Open Google Colab, and mount your Google Drive using:

```
from google.colab import drive
drive.mount('/content/drive')
```

4. Finally, run the notebook cells in order.

3. Technological Challenges:

During the project, we encountered several technical challenges:

- Handling large-scale data:

  The dataset contains millions of flight records, which made traditional tools like Pandas inefficient for processing. To address this, we used PySpark and Spark SQL to perform distributed data processing, enabling efficient computation on large datasets.

- Data preprocessing complexity:

  The raw data was provided as multiple monthly ZIP files, which required extraction, merging, and transformation into a unified format. In addition, we performed extensive data cleaning and feature engineering, such as creating new columns (e.g., delay indicators, departure periods, and seasonal categories) to support further analysis.

- Missing values and inconsistent data:

  The dataset contained missing values and inconsistencies across several columns (e.g., delay times and timestamps). We handled these issues through filtering, type conversion, and conditional transformations to ensure data quality and consistency for analysis.

- Model limitations:

  The machine learning model achieved moderate accuracy (~78%), but its performance was limited by the lack of important real-world features such as weather conditions,

airport congestion, and real-time operational factors. Improving accuracy further would require using variables that directly reflect delay outcomes, which could introduce data leakage. Therefore, we focused on demonstrating the modeling process while emphasizing its limitations and prioritizing data analysis insights.

4. Changes in Technology

We initially considered using Dask for large-scale data processing, since it is designed to parallelize Pandas operations and is relatively easy to use. However, after trying it, we found that Dask was not the best fit for our project. First, Dask works well for medium-scale data, but our dataset contains millions of flight records, and performance was not as stable when running more complex operations such as joins and aggregations. Second, our project also required machine learning. While Dask supports some ML workflows, it is not as integrated as Spark ML (MLlib). Using PySpark allowed us to keep both data processing and modeling within the same framework, which simplified our pipeline. In addition, PySpark provides better support for SQL-based analysis (Spark SQL), which we used extensively in our project. This made querying and aggregating the data much more convenient. Therefore, we chose PySpark over Dask because it is more robust for large-scale data, better integrated with machine learning, and more suitable for end-to-end big data workflows.

Next, we initially considered using a Dataproc (Hadoop/Spark) cluster for this project. However, it required more setup and configuration effort, including cluster management and resource allocation. Given the scope of our project, we chose to use Google Colab with PySpark instead, which provided a simpler and more convenient environment. Colab allowed us to run Spark-based analysis efficiently without dealing with infrastructure overhead, making it a more practical choice for our workflow.

5. Uncovered Aspects from Presentations:

During our presentation, we covered the main components and key findings of the project. However, one important aspect that was not discussed in detail is how we selected features for our analysis and machine learning model.

In this project, feature selection was mainly based on domain understanding and data availability. We selected features that are commonly known to influence flight delays, such as airline, departure time, distance, and delay-related historical information. These features were directly available in the dataset and could be efficiently processed using PySpark.

Core Column Selection

The raw airline dataset contains many fields.
To keep the analysis focused and efficient, we select a subset of core columns related to:

- flight date and schedule
- airline identity
- origin and destination airports
- departure and arrival delays
- cancellation and diversion indicators
- elapsed time and distance
- detailed delay causes

```
core_columns = [
    "Year", "Quarter", "Month", "DayofMonth", "DayOfWeek",
    "FlightDate",

    "Reporting_Airline",

    "Origin", "OriginCityName", "OriginState",
    "Dest", "DestCityName", "DestState",

    "CRSDepTime", "DepTime", "DepDelay", "DepDelayMinutes",
    "CRSArrTime", "ArrTime", "ArrDelay", "ArrDelayMinutes",

    "Cancelled",  "Diverted",

    "CRSElapsedTime", "ActualElapsedTime", "AirTime", "Distance",

    "CarrierDelay", "WeatherDelay", "NASDelay",
    "SecurityDelay", "LateAircraftDelay"
]

df_core = df.select(*core_columns)
print("Selected columns:", len(df_core.columns))
print(df_core.columns)
```

```
Selected columns: 32
['Year', 'Quarter', 'Month', 'DayofMonth', 'DayOfWeek', 'FlightDate', 'Reporting_Airline', 'Origin', 'OriginCityName', 'OriginState', 'Dest', 'DestCityName', 'DestState', 'CRSDepTime', 'DepT
```

In addition, we performed feature engineering to create more meaningful variables. For example, we derived features such as:

- Departure time categories (morning, afternoon, evening, night)

- Seasonal indicators based on month
- Binary delay indicators (e.g., whether a flight is delayed by more than 15 minutes)
- Weekend indicators and long-flight indicators

```python
# Create delay indicators and basic time features
df_clean = df_core \
    .withColumn("is_cancelled", col("Cancelled").cast("int")) \
    .withColumn("is_diverted", col("Diverted").cast("int")) \
    .withColumn("is_dep_delayed", when(col("DepDelayMinutes") > 15, 1).otherwise(0)) \
    .withColumn("is_arr_delayed", when(col("ArrDelayMinutes") > 15, 1).otherwise(0)) \
    .withColumn("CRSDepTime_PADDED", lpad(col("CRSDepTime").cast("string"), 4, "0")) \
    .withColumn("dep_hour", floor(col("CRSDepTime_PADDED").substr(1, 2).cast("int")))
```

```python
# Create seasonal feature
df_clean = df_clean.withColumn(
    "season",
    when(col("Month").isin(12, 1, 2), "Winter")
    .when(col("Month").isin(3, 4, 5), "Spring")
    .when(col("Month").isin(6, 7, 8), "Summer")
    .otherwise("Fall")
)
```

```python
# Create additional features
df_clean = df_clean.withColumn(
    "is_weekend",
    when(col("DayOfWeek").isin(6, 7), 1).otherwise(0)
)

df_clean = df_clean.withColumn(
    "dep_period",
    when((col("dep_hour") >= 5) & (col("dep_hour") < 12), "Morning")
    .when((col("dep_hour") >= 12) & (col("dep_hour") < 17), "Afternoon")
    .when((col("dep_hour") >= 17) & (col("dep_hour") < 21), "Evening")
    .otherwise("Night")
)

df_clean = df_clean.withColumn(
    "is_long_flight",
    when(col("Distance") >= 1500, 1).otherwise(0)
)
```

These engineered features helped simplify the data and capture important patterns for both analysis and prediction.

6. Lessons Learned

From this project, we gained several important insights related to large-scale data processing, data analysis, and machine learning.

First, data preprocessing is a critical step, especially when working with large datasets. Since the data was stored in multiple ZIP files, we needed to extract, clean, and merge the data before any analysis. This process required significant effort but was essential to ensure data quality and consistency.

Second, we learned that PySpark is much more suitable than traditional tools like Pandas for large-scale data processing. While Pandas works well for small datasets, it becomes inefficient when handling millions of records. PySpark enabled us to efficiently process, transform, and analyze large volumes of data.

Third, data visualization plays a key role in understanding complex patterns. Through visualizations, we were able to clearly identify important trends such as differences in delay performance across airlines, delay accumulation throughout the day, and variations over time.

Fourth, we learned that flight delays are influenced by multiple interacting factors, rather than a single cause. Factors such as airline operations, airport congestion, and time of day all contribute to delay patterns, making the problem inherently complex.

Finally, we found that machine learning performance is strongly limited by the available data. Although we built a prediction model, the accuracy remained moderate (around 60–78%) due to missing critical features such as weather conditions and real-time operational data. This

highlights that having the right data is more important than simply using more complex models, and that real-world prediction tasks often involve unavoidable limitations.

7. Future Improvements

If more time and resources were available, several improvements could be made to enhance this project.

First, more advanced machine learning models such as Random Forest, Gradient Boosting, or XGBoost could be explored. In this project, we used Logistic Regression due to its simplicity and efficiency in a big data environment. However, flight delays are influenced by complex and non-linear interactions between multiple factors, which more advanced models may capture more effectively.

Second, the dataset may suffer from class imbalance, as non-delayed flights are more common than delayed ones. This imbalance can bias the model toward the majority class. In future work, techniques such as resampling or class balancing methods (e.g., SMOTE) could be applied to improve model performance.

Third, incorporating external data sources would significantly improve the quality of analysis and prediction. Currently, the dataset lacks important real-world factors such as weather conditions, airport congestion, and real-time traffic information, which are known to strongly influence flight delays.

In addition, further improvements could be made in feature engineering. While we created several useful features (e.g., time periods, seasons, and delay indicators), more advanced feature construction could help better represent complex patterns in the data.

Finally, this project could be extended into a real-time prediction system. For example, a system or API could be developed to allow users to input flight information and receive delay predictions in real time. This would make the project more practical and closer to real-world applications.

8. Data Sources, and Results:

Code Repository: https://github.com/TianqiZhen426/Big-Data_2026Spring_Project/tree/main

Data Sources:

U.S. Bureau of Transportation Statistics

How to Download Dataset:

- Click on this link to find the datasets https://www.transtats.bts.gov/Tables.asp?QO_VQ=EFD&QO_anzr=Nv4yv0r%FDb0-gvzr%FDcr4s14zn0pr%FDQn6n&QO_fu146_anzr=b0-gvzr
- Click “Download” on the second dataset “Reporting Carrier On-Time Performance (1987-present)” or use this link https://www.transtats.bts.gov/Fields.asp?gnoyr_VQ=FGJ
- Select year and attributes and click on “Download”

Data Selection

In this project, we selected data from the year 2024 for analysis. The original BTS dataset is extremely large, so we restricted our analysis to one year to keep the project manageable. Each monthly file expands to over 250MB, resulting in more than 3GB of data for the full year. This still provides a sufficiently large dataset for meaningful analysis while keeping computational costs reasonable.

Download link:

https://drive.google.com/file/d/1iXESKbrhonsKsb8BJKz7R4o4HPaEYRPj/view?usp=sharing

The dataset includes records from January to December 2024, covering a full year of flight operations. This allows us to capture seasonal patterns and variations in flight delays.

Results and explain

Data Cleaning result:

Cleaned data table

The cleaning process converted the raw BTS dataset into a reliable, structured dataset suitable for Spark SQL analysis, visualization, and machine learning prediction. The following table is the cleaned core columns, it include the data that we will use for the data analysis and predictive model.

```
print("Sample cleaned data:")
df_clean.show(5, truncate=False)

Sample cleaned data:
+----+-------+-----+---------+---------+-----------+---------------+------+---------------+----------+----+---------------+---------+-----------+---------+---------+---------+---------+
|Year|Quarter|Month|DayofMonth|DayOfWeek|FlightDate|Reporting_Airline|Origin|OriginCityName|OriginState|Dest|DestCityName|DestState|CRSDepTime|DepTime|DepDelay|DepDelayMinutes|CRSArrTime|ArrTime|
+----+-------+-----+---------+---------+-----------+---------------+------+---------------+----------+----+---------------+---------+-----------+---------+---------+---------+---------+
|2024|1      |1    |8        |1        |2024-01-08|9E             |LGA   |New York, NY   |NY        |OMA |Omaha, NE      |NE       |856        |851      |-5.0     |0.0      |1135     |1124   |
|2024|1      |1    |9        |2        |2024-01-09|9E             |LGA   |New York, NY   |NY        |OMA |Omaha, NE      |NE       |856        |851      |-5.0     |0.0      |1135     |1107   |
|2024|1      |1    |10       |3        |2024-01-10|9E             |LGA   |New York, NY   |NY        |OMA |Omaha, NE      |NE       |856        |850      |-6.0     |0.0      |1135     |1110   |
|2024|1      |1    |11       |4        |2024-01-11|9E             |LGA   |New York, NY   |NY        |OMA |Omaha, NE      |NE       |856        |919      |23.0     |23.0     |1135     |1202   |
|2024|1      |1    |12       |5        |2024-01-12|9E             |LGA   |New York, NY   |NY        |OMA |Omaha, NE      |NE       |856        |851      |-5.0     |0.0      |1135     |1137   |
+----+-------+-----+---------+---------+-----------+---------------+------+---------------+----------+----+---------------+---------+-----------+---------+---------+---------+---------+
only showing top 5 rows
```

```
print("Sample cleaned data:")
df_clean.show(5, truncate=False)

...
+-------+--------+---------------+---------+--------+---------------+---------------+-------+--------+-----------+---------+---------+---------+---------+---------+---------+---------+
:|ArrTime|ArrDelay|ArrDelayMinutes|Cancelled|Diverted|CRSElapsedTime|ActualElapsedTime|AirTime|Distance|CarrierDelay|WeatherDelay|NASDelay|SecurityDelay|LateAircraftDelay|is_cancelled|is_diverted|
+-------+--------+---------------+---------+--------+---------------+---------------+-------+--------+-----------+---------+---------+---------+---------+---------+---------+---------+
|1124   |-11.0   |0.0            |0.0      |0.0     |219.0          |213.0          |184.0  |1148.0  |0.0        |0.0      |0.0      |0.0      |0.0      |0        |0        |
|1107   |-28.0   |0.0            |0.0      |0.0     |219.0          |196.0          |168.0  |1148.0  |0.0        |0.0      |0.0      |0.0      |0.0      |0        |0        |
|1110   |-25.0   |0.0            |0.0      |0.0     |219.0          |200.0          |177.0  |1148.0  |0.0        |0.0      |0.0      |0.0      |0.0      |0        |0        |
|1202   |27.0    |27.0           |0.0      |0.0     |219.0          |223.0          |188.0  |1148.0  |0.0        |0.0      |4.0      |0.0      |23.0     |0        |0        |
|1137   |2.0     |2.0            |0.0      |0.0     |219.0          |226.0          |185.0  |1148.0  |0.0        |0.0      |0.0      |0.0      |0.0      |0        |0        |
+-------+--------+---------------+---------+--------+---------------+---------------+-------+--------+-----------+---------+---------+---------+---------+---------+---------+---------+
```

```
print("Sample cleaned data:")
df_clean.show(5, truncate=False)

+---+-----------+-----------+-------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+
:e|CarrierDelay|WeatherDelay|NASDelay|SecurityDelay|LateAircraftDelay|is_cancelled|is_diverted|is_dep_delayed|is_arr_delayed|CRSDepTime_PADDED|dep_hour|season|is_weekend|dep_period|is_long_flight|
+---+-----------+-----------+-------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+
|0.0|0.0|0.0|0.0|0.0|0|0|0|0|0|0856|8|Winter|0|Morning|0|
|0.0|0.0|0.0|0.0|0.0|0|0|0|0|0|0856|8|Winter|0|Morning|0|
|0.0|0.0|0.0|0.0|0.0|0|0|0|0|0|0856|8|Winter|0|Morning|0|
|0.0|0.0|4.0|0.0|23.0|0|0|1|1|0856|8|Winter|0|Morning|0|
|0.0|0.0|0.0|0.0|0.0|0|0|0|0|0856|8|Winter|0|Morning|0|
+---+-----------+-----------+-------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+-----------+
```

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This figure shows a sample of the cleaned dataset after preprocessing. The raw BTS data was transformed into a structured format suitable for analysis and machine learning.

The cleaned dataset includes both original features (such as flight date, airline, departure time, and delay values) and newly engineered features. These engineered features include variables such as delay indicators (is_dep_delayed, is_arr_delayed), time-based features (dep_hour, dep_period), seasonal categories, and flight characteristics (is_long_flight, is_weekend).

This preprocessing step ensures that the data is consistent, complete, and ready for large-scale analysis using Spark SQL, as well as for building predictive models.

Data Analysis results:

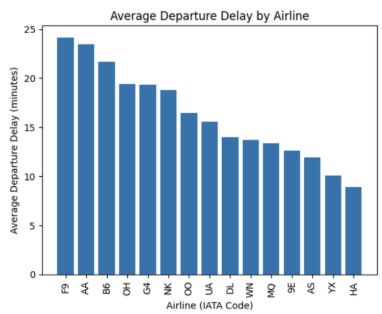
The following figures illustrate the results obtained from our Spark SQL analysis, along with the key insights identified in each section.

Airline Performance Analysis

- What is the average delay time for each airline?
- Which airlines have the highest delay times (Top 10)?

```
+-----------------+---------+
|Reporting_Airline|avg_delay|
+-----------------+---------+
|               F9|    24.17|
|               AA|    23.49|
|               B6|    21.68|
|               OH|    19.43|
|               G4|    19.37|
|               NK|     18.8|
|               OO|    16.42|
|               UA|    15.56|
|               DL|    14.01|
|               WN|    13.71|
+-----------------+---------+
only showing top 10 rows
```

...

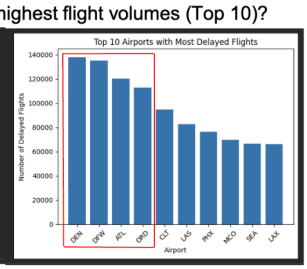


Key Finding:
The average delay duration for the top 10 airlines is significantly higher, and there are substantial disparities in delay performance among these airline company. Some airlines are notably more prone to delays than others.

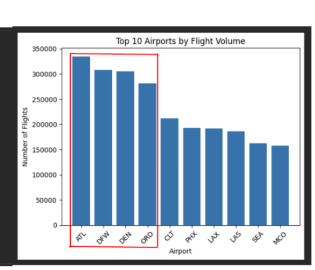
Airport Analysis

- Which airports experience the most delays (Top 10)?
- Which airports have the highest flight volumes (Top 10)?

```
+------+------------+
|Origin|delay_count|
+------+------------+
|   DEN|      138055|
|   DFW|      135243|
|   ATL|      120438|
|   ORD|      113048|
|   CLT|       94881|
|   LAS|       82762|
|   PHX|       76526|
|   MCO|       69560|
|   SEA|       66475|
|   LAX|       66303|
+------+------------+
only showing top 10 rows
```




```
+------+-------------+
|Origin|flight_count|
+------+-------------+
|   ATL|       334706|
|   DFW|       308078|
|   DEN|       305201|
|   ORD|       280821|
|   CLT|       211657|
|   PHX|       192685|
|   LAX|       191946|
|   LAS|       186526|
|   SEA|       162268|
|   MCO|       157163|
+------+-------------+
only showing top 10 rows
```

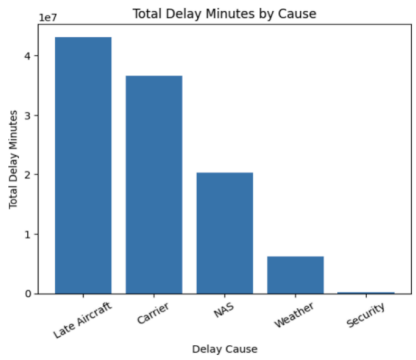


Key Finding:
The airports with the most delayed flights are also among the busiest airports, including DEN, DFW, ATL, and ORD. Higher traffic volume generally leads to more congestion and delays, although airport-specific factors such as weather and operational efficiency also affect performance.

Delay Cause Analysis

- What are the most common causes of delays?

```
...  +----------+---------+----------+--------+-------------+
     |   Carrier|  Weather|       NAS|Security|Late Aircraft|
     +----------+---------+----------+--------+-------------+
     |3.6621557E7|6267029.0|2.0242996E7|183950.0|  4.3136577E7|
     +----------+---------+----------+--------+-------------+
```

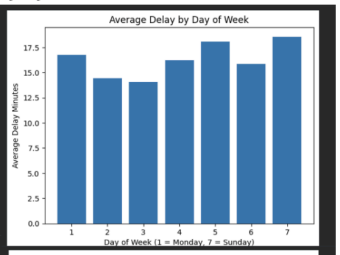


Key Finding:
Weather is not the biggest factor affecting the delays. Late aircraft arrivals are the primary driver of delays, showing that delay propagation is the major operational challenge for airlines.

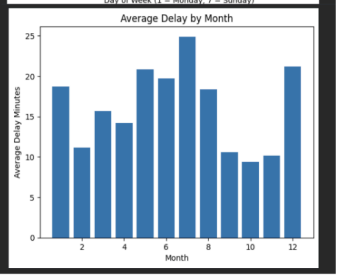
Temporal Analysis

- How do delays vary by day of the week?
- How do delays vary by month?

```
+---------+---------+
|DayOfWeek|avg_delay|
+---------+---------+
|        1|    16.76|
|        2|    14.42|
|        3|    14.08|
|        4|    16.21|
|        5|    18.08|
|        6|    15.87|
|        7|    18.56|
+---------+---------+
```




```
..  +-----+---------+
    |Month|avg_delay|
    +-----+---------+
    |    1|    18.75|
    |    2|    11.18|
    |    3|    15.68|
    |    4|    14.18|
    |    5|    20.85|
    |    6|    19.69|
    |    7|     24.9|
    |    8|    18.39|
    |    9|    10.56|
    |   10|     9.39|
    |   11|    10.17|
    |   12|    21.18|
    +-----+---------+
```

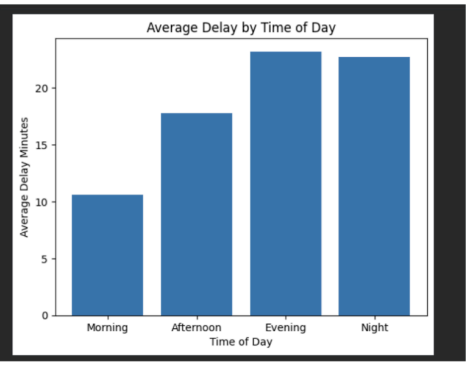


- Day-of-week differences are relatively small, suggesting weekday alone is not a major delay driver.
- Monthly differences are more noticeable, with higher delays in May, June, July, Dec and Jan.
- This indicates that seasonal travel demand, weather conditions, and holiday periods may have strong effects.

Departure Time Analysis

- How do delays vary by time of day (morning, afternoon, evening, night)?

```
+----------+---------+
|dep_period|avg_delay|
+----------+---------+
|   Evening|     23.2|
|   Morning|    10.59|
| Afternoon|    17.79|
|     Night|    22.75|
+----------+---------+
```



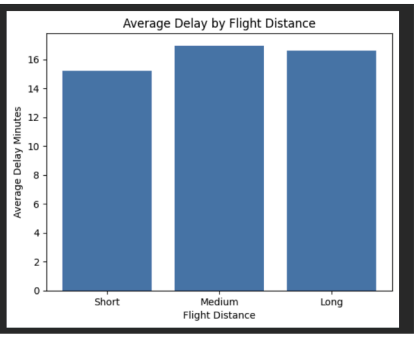
Key Finding:
Flight delays increase steadily throughout the day, with the highest delays in the night. Later departures are more delay-prone because delays accumulate and propagate across daily flight schedules.

NYU

Distance Analysis

- What is the relationship between flight distance and delay time?

```
+--------------+---------+
|distance_group|avg_delay|
+--------------+---------+
|        Medium|    16.96|
|          Long|    16.59|
|         Short|    15.23|
+--------------+---------+
```



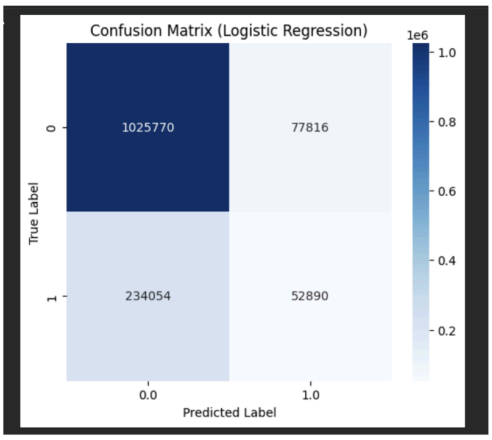
Average delays are similar across short, medium, and long-distance flights. Flight distance does not appear to be a major driver of delays.

Prediction Model result:

Model Result

```
Logistic Regression Results
AUC: 0.67
Accuracy: 0.78
F1 Score: 0.74
```

We used a logistic regression model to predict flight delays.
The accuracy is around 60 to 78 percent, which is not very high.
This is mainly because important factors such as weather and airport congestion are missing.
Therefore, predicting flight delays is a challenging problem.



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